Smaller Coefficients for a Quartic in 214 Variables

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Abstract

An invertible rational rescaling of six witness coordinates reduces the coefficient growth of the specified 214-variable quartic family. The original coefficient height is $149^{4N+2}+1$ for $N \geq 53$. The new height is exactly $31581660^{28}149^{2N}$ for $N \geq 57$ and is strictly smaller than the original height for every $N \geq 53$. The rational input projection is unchanged for each parameter. All 188 quadratic equations and all 2919 quartic coefficient profiles are checked exactly. The argument is algebraic: it does not verify the original arithmetic foundations or make the sufficient defining threshold effective.

1 The source and the statement

The source polynomial is the sum of squares of 188 primitive integer quadratic equations in 214 rational coordinates. Its first coordinate is the distinguished input. Write the fixed coefficient as

$$A_N = 149^{2N+1}, \qquad P_N(X_1, \ldots, X_{214}) = \sum_{i=1}^{188} r_i(A_N, \boldsymbol{X})^2.$$

The exact source version is the family retained in the complete Version 17 manuscript at commit

`1f7e1f1fb0438feda3c9f3da9cc16982126aeeed`.

Its published coefficient file has Git blob identifier

`f6a32cc2f7ebcc37a952c5ce737169608cfe27c2`.

The accompanying source table is reconstructed from the printed equations. Its sum of squares reproduces that exact published coefficient file; no assertion of byte identity with the source's separate quadratic table is needed. The source and all new tables are retained in

`computation/quartic/balanced214_v24/`.

Set

$$K = 31581660^{28}, \qquad J = 5898050446009651968, \qquad H_0 = (JK)^2.$$

Coefficient height means the largest absolute integer coefficient after substitution of the fixed parameter. All coordinate counts include the input.

Theorem 1.1. *There is a completely specified integer quartic family $\widetilde{P}_N \in \mathbb{Z}[Y_1, \ldots, Y_{214}]$, for integers $N \geq 1$, with these properties.*

1. *For every $x \in \mathbb{Q}$ and every $N \geq 1$,*

$$\exists \boldsymbol{y} \in \mathbb{Q}^{213}\ P_N(x, \boldsymbol{y}) = 0 \quad \Longleftrightarrow \quad \exists \boldsymbol{z} \in \mathbb{Q}^{213}\ \widetilde{P}_N(x, \boldsymbol{z}) = 0.$$

2. *Each member has degree exactly four, integer content one, and exactly 2919 nonzero coordinate monomials. It is a sum of 188 primitive quadratic squares.*

3. For every $N \geq 1$,

$$H(\widetilde{P}_N) = \max\{H_0, K149^{2N}\}.$$

For $N \geq 57$ this is $K149^{2N}$. For every $N \geq 53$,

$$H(\widetilde{P}_N) < 149^{4N+2} + 1 = H(P_N).$$

The sufficient defining threshold of the source is not evaluated here. Consequently, if its integer-definability theorem is used with its stated premises and a fixed ineffective threshold N_0, exactly the same N_0 works for the new family. Neither 53 nor 57 is asserted to be a sufficient defining threshold. This is a coefficient improvement of that source, not a new variable-count result or an independent proof of its global arithmetic theorem.

2 Rational rescaling and row normalization

Put $C = 149^N$, so $A_N = 149C^2$, and let

$$I = \{90, 91, 93, 117, 118, 120\}.$$

Make the coordinate substitution

$$X_j = \begin{cases} CY_j, & j \in I, \\ Y_j, & j \notin I. \end{cases} \tag{1}$$

This is an invertible rational linear map for every nonzero rational C; it fixes $X_1 = Y_1$.

For the source row numbers, define

$$m_{35} = 2, \qquad m_{36} = m_{37} = m_{38} = m_{60} = m_{63} = m_{65} = 1,$$

and $m_i = 0$ for all other rows. The new primitive rows are

$$\widetilde{r}_i(C, \boldsymbol{Y}) = \varepsilon_i C^{-m_i} r_i(149C^2, \boldsymbol{X}(C, \boldsymbol{Y})), \qquad \varepsilon_i \in \{1, -1\}. \tag{2}$$

The sign is chosen so the first coefficient in the prescribed monomial order is positive; it has no effect on the sum of squares. Each numerator in (2) is divisible by the stated power of C in $\mathbb{Z}[C, \boldsymbol{Y}]$. The quotient has integer content one, degree at most two in $\boldsymbol{Y}$, and degree at most one in C.

These statements can be checked term by term without evaluating any large parameter. A source term $aA^e X_{j_1} \cdots X_{j_k}$ becomes

$$a149^e C^{2e+\#\{\ell : j_\ell \in I\}} Y_{j_1} \cdots Y_{j_k}.$$

For each of the 188 rows the minimum exponent is m_i, and the difference between maximum and minimum exponents is at most one. The integer coefficients after division have greatest common divisor one. The supplied checker verifies the displayed identity for every term and every row. No coordinate monomial in an individual row has both a constant and a linear coefficient in C. If g_0, g_1 are the greatest common divisors of the constant and linear coefficient groups of a row, the checker gives $\gcd(g_0, g_1) = \gcd(g_0, 149) = 1$. Thus every specialization $C = 149^N$ remains primitive over $\mathbb{Z}$. Fifty-two rows change; the remaining 136 retain their coefficient vectors up to the harmless chosen signs.

Define

$$\widetilde{P}_N(\boldsymbol{Y}) = \sum_{i=1}^{188} \widetilde{r}_i(149^N, \boldsymbol{Y})^2.$$

Over $\mathbb{Q}$, a sum of squares vanishes exactly when each summand vanishes. All the row multipliers in (2) are nonzero, and the coordinate map (1) is bijective. This proves both directions of the projection assertion, on the entire rational solution set. No witness coordinate is divided out; zero witness values remain allowed. No local or global arithmetic existence assertion is added.

3 Exact coefficient height

Collect the expansion by its coordinate monomials M. Write

$$\widetilde{P}(C, \boldsymbol{Y}) = \sum_{M} p_M(C) M(\boldsymbol{Y}).$$

The exact expansion has 2924 terms before equal coordinate monomials are collected, and 2919 after collection. The coefficient types are

Nonzero powers of C in p_M	Number of coordinate monomials
$\{0\}$	2508
$\{1\}$	331
$\{2\}$	75
$\{0, 2\}$	5

In each of the five mixed profiles the two coefficients have the same sign. Therefore no coordinate monomial disappears when $C > 0$. Every one of the 214 coordinates occurs. The coefficient of Y_1^4 is exactly one for every C, proving degree four and content one.

The two coefficients

$$p_{Y_{112}^4}(C) = H_0, \qquad p_{Y_{109}^2 Y_{117}^2}(C) = KC^2 \tag{3}$$

will attain the height. The following elementary observation makes the check for all parameters finite.

Lemma 3.1. *Let $H_0, K > 0$ and $C > 0$.*

1. *If $a^2 \le H_0 K$, then $|a|C \le \max(H_0, KC^2)$.*
2. *If $|a_0|K + |a_2|H_0 \le H_0 K$, then $|a_0 + a_2 C^2| \le \max(H_0, KC^2)$.*

Proof. The geometric mean of H_0 and KC^2 is no larger than their maximum, which proves the first assertion. For the second, put $u = |a_0|/H_0$ and $v = |a_2|/K$. Then $u, v \ge 0$ and $u + v \le 1$. The triangle inequality gives

$$|a_0 + a_2 C^2| \le uH_0 + vKC^2 \le \max(H_0, KC^2).$$

□

For each of the 331 linear profiles the exact integer calculation gives $a^2 \le H_0 K$. For each of the remaining 2588 profiles it gives $|a_0|K + |a_2|H_0 \le H_0 K$, with a missing coefficient taken to be zero. Every coefficient is therefore bounded by the claimed maximum, and (3) proves equality. The verification concerns every coefficient, not just an upper bound from a circuit norm.

To locate the integer threshold, the checker verifies

$$149^{56} < J31581660^{14} < 149^{57}.$$

Since $H_0/K = J^2 K$, it follows that $KC^2 > H_0$ for every $C = 149^N$ with $N \ge 57$, and that $H_0 > KC^2$ for $N \le 56$. The strict improvement over the source follows from the two exact inequalities

$$H_0 < 149^{214} + 1, \qquad K < 149^{108}.$$

For $N \ge 53$, the first is bounded by $149^{4N+2} + 1$, and the second gives $K149^{2N} < 149^{4N+2}$. These prove the height assertions of Theorem 1.1.

For completeness the source height is also rechecked: its unique quadratic profile in A is $A^2 + 1$. Every other profile has degree at most one and absolute value strictly smaller at $A = 149^{107}$. After division by A^2, the corresponding absolute-value bounds decrease as A increases. This establishes $H(P_N) = 149^{4N+2} + 1$ for all $N \ge 53$, independently of the new coefficient calculation.

4 Reproducibility and audit scope

The accompanying standard-library producer writes the full new quadratic and quartic tables. A separate checker imports no producer and expands squares using unordered term pairs rather than the producer's ordered double loop. It performs the following finite checks: source polynomial identity, expansion of all reconstructed source rows, every row transformation, all new quartic coefficients, the exact support and content, all height envelope inequalities, and both parameter thresholds. A deliberately corrupted transformed coefficient is rejected.

The reconstructed source polynomial has 246349 bytes and SHA-256

`9f461531df52ac73e02cf7a4ce270422715642c0d3112952cad4968d9a7e24ea`.

Its Git blob identifier matches the live repository metadata at the pinned commit. This binds the numerical source used here to the actual published polynomial, not merely to a count or an old review verdict. Coordinate names and the reconstructed equations are distributed beside the checker. The named indices in the proof are the original source indices, so no coordinate renumbering is implicit.

These algebraic checks do not establish the source's elliptic rank, prime-pattern, pole-realization or contact-height theorems. In particular they do not supply complete independent acceptance of the assembled arithmetic proof. The producer, checker and written argument were developed in one research session; distinct implementations are not presented as non-author review. The statement proved here is exact algebraic projection equivalence and a coefficient improvement, with the original arithmetic status and quantifier order left unchanged.